



RealSense™ Product Family

D400 Series

Specification Update

Revision 045

June 2026

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Revision History

Revision Number	Description	Revision Date	Comment
001	Firmware 5.8.15 Release	February 2018	
002	Firmware 5.9.2 Release	March 2018	
003	Firmware 5.9.9 Release	April 2018	
004	Firmware 5.9.11 Release	May 2018	
005	Firmware 5.9.13 Release	June 2018	
006	Firmware 5.9.14	July 2018	
007	Firmware 5.10.3	August 2018	
008	Firmware 5.10.6	October 2018	
009	Firmware 5.10.13	November 2018	
010	Firmware 5.11.1	January 2019	
011	Firmware 5.11.1.100	February 2019	
012	Firmware 5.11.4	February 2019	
014	Firmware 5.11.6.250	June 2019	
015	Firmware 5.11.11.100	August 2019	
016	Firmware 5.11.15.0	September 2019	
017	Firmware 5.12.0	November 2019	
018	Firmware 5.12.1	December 2019	
019	Firmware 5.12.2.100	January 2020	
020	Firmware 5.12.3.0	February 2020	
021	Firmware 5.12.5.0	June 2020	
022	Firmware 5.12.6.0	July 2020	
023	Firmware 05.12.07.100	August 2020	
024	Firmware 05.12.08.200	October 2020	
025	Firmware 05.12.09.00	November 2020	
026	Firmware 05.12.10.00	January 2021	
027	Firmware 05.12.11.00	February 2021	
028	Firmware 05.12.12.100	March 2021	
029	Firmware 05.12.13.50	May 2021	
030	Firmware 05.12.14.50	June 2021	
031	Firmware 05.12.15.50	October 2021	
032	Firmware 05.13.0.50	November 2021	

033	Firmware 05.14.0.0	December 2022	
034	Firmware 05.15.0.2	June 2023	
035, 036	Firmware 05.15.1.0	November 2023	
037	Firmware 05.16.0.1	April 2024	
038	Firmware 05.15.1.55	September 2024	Check "Table 2 2. Specification Changes"
039	Firmware 05.16.3.0	February 2025	Check "Table 2 2. Specification Changes"
040	Firmware 5.17.0.9	July 2025	
041	Firmware 5.17.0.9	July 2025	Updated Errata
042	Firmware 5.17.0.10	July 2025	
043	Firmware 5.17.0.213	March 2026	D436
044	Firmware 5.17.2.201	April 2026	Vision Processor (VP) D4 Board V6 + D401 GMSL
045	Firmware 5.17.0.214	June 2026	D436
045	Firmware 5.17.3.10	June 2026	D435, D455, D457

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1 Preface

This document lists the firmware versions supporting the RealSense™ Product family D500 Series.

1.1 Terminology

Errata are design defects or errors. These may cause behavior to deviate from published specifications.

Specification Changes are modifications to the current published specifications. These changes will be incorporated in any new release of the specifications.

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2 Summary Table of Changes

The following tables indicate the errata and specification changes which apply to the RealSense Products. may fix some of the errata in a future release of the component and account for the other outstanding issues through documentation or specification changes as noted.

2.1 Codes Used in Summary Tables

Status

Open:	In engineering assessment
Fixed:	This erratum has been previously fixed
No Fix:	There are no plans to fix this erratum

Table 2-1. Errata Summary Table

Number	Status	Errata
RSDSO-7194	Fixed in Windows 10* RS4	Windows driver [mksrv.sys] may crash with D400 series cameras in stress test condition
RSDSO-7755	Fixed in Firmware 5.9.9	Temporary stream hang observed on disabling Auto Exposure (AE) at 1280X720 resolution after any previous 90 FPS stream
RSDSO-7798	Fixed in Firmware 5.9.2	RGB at 60 FPS may not have the right exposure set when exposure is equal/less than -2
RSDSO-7849	Fixed in Firmware 5.9.9	ROI based depth streaming immediately after change of IR projector power may result in a stream hang
RSDSO-7854	Fixed in Firmware 5.9.13	Depth Stream hang when system resumes from Sleep (S3)
RSDSO-7976	Fixed in Firmware 5.9.2	D400 Series camera is not recognized after reboot on Linux
RSDSO-8007	Fixed in Windows Driver DMFT_5.160.1.5+	Firmware updates via DFU fails when firmware update limit is reached
RSDSO-8328	Fixed in Firmware 5.9.13	Metadata attribute "Trigger" indicating Depth to Color synchronization may not have correct value
RSDSO-8461	Not a RealSense bug (Requires Chrome* fix)	D400 Series Windows UWP driver does not work with Chrome browser
RSDSO-8467	Fixed in LibRealSense2.10.1	Left Imager UYVY format displays green image
RSDSO-8538	Fixed in Firmware 5.9.11	Color correction parameters are not updated correctly
RSDSO-8565	No Fix (expected as per current design)	Infrared speckles on color image from D415 and D435 cameras
RSDSO-6804 RSDSO-8681	No Fix	D400 Series cameras intermittently enumerated as USB2 device on unplug/plug
RSDSO-9006	Fixed in Firmware 5.9.11	Frame rate does not change when manual exposure value is changed
RSDSO-9074	Closed. This is USB2 Bandwidth related	Simultaneous streaming Depth, Imager and Color may result in data stream hang when camera is connected through USB2

Summary Table of Changes

Number	Status	Errata
RSDSO-9094	Fixed in Firmware 5.9.11	Specific controls values missing in frames metadata
RSDSO-9153	Fixed in Firmware 5.9.11	D400 series camera fails to be recognized on system reboot when connected through USB3
RSDSO-9224	Fixed in Firmware 5.9.13	IR Projector pattern flicker when streaming through USB2 connection
RSDSO-9228	Fixed in Firmware 5.9.13	D400 series camera disconnects on resume from system sleep when connected through USB2
RSDSO-9240	Fixed in Firmware 5.9.13	D400 Series camera fails to be recognized on system reboot when connected through USB2
RSDSO-9478	Fixed in Firmware 5.10.6	Image Flicker when Auto Exposure (AE) is enabled
RSDSO-9501	Fixed in Windows 10* RS5	Camera is not functional after HLK Sensor test when connected through USB2
RSDSO-9546	Fixed in Firmware 5.9.14	IR projector pattern flicker when streaming at 1280X720, 4 FPS and connected through USB2
RSDSO-9556	Fixed in Firmware 5.10.3	Camera stuck after streaming start-stop at Low FPS for few times
RSDSO-9645	Fixed in Firmware 5.10.3	Darker depth frame when changing depth exposure from [165760 - 165780] and connected through USB2
RSDSO-10002	Fixed in Firmware 5.10.3	Calibration tables may get corrupted during power on and off cycles
RSDSO-10011	No Fix	Auto Exposure (AE) for Color is not optimized for bright sunlight
RSDSO-10428	Fixed in Firmware 5.11.4	IR Image is black in Auto Exposure (AE) mode with sudden exposure to light
RSDSO-10431	Fixed in Firmware 5.11.1	IR image may flicker in outdoor sunlight when using Auto Exposure (AE) with default set point
RSDSO-10503	No Fix	Low fill rate in outdoor environment using Auto Exposure (AE)
RSDSO-10603	Fixed in Firmware 5.12.3.0	Unable to set the Depth Exposure Time < 70 usec through USB2
RSDSO-10777	Fixed in LibRealSense	D435i - Buffer overflow on repeated start/stop
RSDSO-10674	Fixed in LibRealSense	D435i – The first (cold) start of IMU sensors in LibRealSense on Linux takes ~4 sec
RSDSO-10920	Fixed in LibRealSense	First frames not received on metadata test
RSDSO-11041	Fixed in LibRealSense	D435i – Unreasonably large accelerometer reading in Windows 10
RSDSO-10591	No Fix	In Multi Camera mode, sporadic and inconsistent frame drops and streaming halt
RSDSO-11040	Fixed in Firmware 5.11.4	D435 – Depth/IR corrupted image when streaming multi stream and RGB exposure is < -6
RSDSO-11042	Fixed in Firmware 5.11.11.100. LibRealSense 2.20.0 required.	D430 – RealSense Viewer errors out post reboot after using the “Hand” preset.
RSDSO-12586	Fixed in Firmware 5.11.11.100	D415, D435/D435i – RGB camera not available after FW Update process using DFU
RSDSO-12587 RSDSO-12814	Fixed in Firmware 5.11.6.250	D435i – Upgrading to FW version 5.11.6.200 causes corrupted calibration table. Note: Due to this issue this FW was removed.
RSDSO-10229 RSDSO-13386	Fixed in Firmware 5.11.15.0	Camera fail after start, stop of random profiles
RSDSO-13540	Fixed in Firmware 5.11.15.0	D415 – RealSense™ Self-Calibration doesn't converge using D415 camera

Summary Table of Changes

Number	Status	Errata
RSDSO-13546	Fixed in Firmware 5.12.0	D435i – IMU frame drops
RSDSO-13554	Fixed in Firmware 5.11.15.0	D420 – Camera will stop working after FW update
RSDSO-12578	Fixed in Firmware 5.12.3.0	Device not recognized in device manager after machine reboot
RSDSO-14309	Fixed in LibRealSense	Scale error after Tare calibration (RealSense™ Self-Calibration) was significantly degraded
RSDSO-14530	Fixed in LibRealSense	Frame drops when running on Ubuntu 18.04 with Kernel 5
RSDSO-14525	Fixed in LibRealSense	D435i – Camera not released to idle state on Ubuntu 18.04 with Kernel 5
RSDSO-14517	Fixed in Firmware 5.12.5.0	10 FPS not supported on Depth/IR 848x480 when connected through USB2
RSDSO-14455	Fixed in Windows 10 RS5	Frames stop arriving after few minutes on Windows 10 RS4
RSDSO-13306	Fixed in Firmware 5.12.5.0	Frames stop arriving after few frames in Color/Depth stream due to unexpected halt on Windows 10 RS4
RSDSO-14512	Fixed in Firmware 5.12.6.0	D435i – Failed to start IMU stream after resetting device
RSDSO-14526	Fixed in Firmware 5.12.5.0	D435i – Sending HW_reset immediately after stop streaming causes the camera to boot into invalid state
RSDSO-14558	Fixed in Firmware 5.12.5.0	Fails USB enumeration during PC restart
RSDSO-14499	Fixed in Firmware 05.12.09.00	Config error when stopping Color stream on Windows 10 RS4
RSDSO-14349	Fixed in LibRealSense	FW Update ID mismatch before and after entering DFU
RSDSO-14337	Fixed in Firmware 5.12.5.0	IR Frames stop arriving after several camera start/stop
RSDSO-14308	No Fix	D415 – Self-Calibration/Tare fails to converge 20% of the time
RSDSO-14978	Fixed in Firmware 5.12.11.0	D455 – Depth/IR frame drops
RSDSO-16289	Fixed in Firmware 5.12.11.0	When laser emitter is enabled via Python, laser turns on and off continuously
RSDSO-16765	Fixed in Firmware 5.12.13.50	Frames not arrived with 300fps
RSDSO-17659	Fixed in Firmware 5.13.0.50	D415 Depth Image fails upon start/stop cycling
RSDSO-17393	Fixed in Firmware 5.13.0.50	Cannot Reset Auto Exp/Gain Limit back to default value 0
RSDSO-17183	Fixed in Firmware 5.13.0.50	D430 depth fill rate is not good with high accuracy preset under 640x480
RSDSO-16964	Fixed in Firmware 5.13.0.50	Emitter_on_off stops when RGB camera stops
RSDSO-17567	Fixed in Firmware 5.13.0.50	RealSense Viewer Crashes after Surface Pro 7+ sleep

Summary Table of Changes

Number	Status	Errata
RSDSO-18928	No Fix, Downgrade is not allowed	New D455 cameras can't downgrade FW to <= 5.12.15.50
RSDSO-19055	No Fix	D450 + D4 ASIC board with FW version 5.12.7.100 cannot be updated
RSDSO-17832	No Fix	D455 & D415: Focal Length on chip calibration does not work with USB2
RSDSO-20061	No Fix	D455/D457: AE issue when streaming at 90fps with laser off and AE enabled
RSDSO-19114	No Fix	D45x Z-accuracy out of Spec at high temp
RSDSO-14986	Fixed in Firmware 5.14.0.0	D455/D457 HW timestamp delta is 4.8FPS and not 5FP
RSDSO-17488	Fixed in release 2.56.1	D405 High UV error when aspect ratio is different than 16/9 (HD)
RSDSO-19269	Fixed in release 2.56.4	D435 RGB enumeration failure after a few HW reset
RSDSO-19297	Fixed in release 2.56.4	D457 camera in recovery mode cannot be recognized by the Viewer
RSDSO-20426	Fixed in Fw v5.17.0.10	Corrupted frames when streaming @ 60 FPS in AE - D455x and D456
RSDEV-3787	No Fix	[D436] Potential RGB degraded image quality
RSDEV-4648	No Fix	[D436] Over high temperature there's degradation in the RGB MTF mainly at the corners
RSDSO-19836	Open	D421: high-pitched sound heard when streaming depth with laser
RSDSO-19674	Fixed	Corrupted depth and IR frames in various streaming scenarios
RSDSO-14504	Open	D435i – Frame drops in gyro stream on Windows
RSDSO-19117	Open	D457 and D456 Aux cap pop up
RSDSO-19280	Open	Self-Calibration: 'NEW' button doesn't show the updated calibration on RealSense Viewer
RSDSO-19017	Open	Up to 80% dropped frames in several builds of Windows 11
RSDSO-20056	Open	D421: Laser is flickering at low exposure values (<1ms)
RSDSO-19865	Open	Laser power change during streaming doesn't work
RSDSO-20018	Open	[D45x] Frames gets corrupted while streaming without laser and the lighting changes
RSDSO-20384	Fixed	[D45x] RGB Manual exposure at 60FPS doesn't work at certain range
RSDSO-20529 / RSDSO-16147	Open	[D45x/D436] No sync between color and depth at 5FPS/6FPS
RSDSO-20465 / RSDSO-20572	Open	[D436] Sequential frame drops when streaming color and depth at high resolution @30 FPS
RSDSO-20978	Open	[VP D4 Board V6 + D401 GMSL]: OEM calibration is supported via USB only
RSDSO-21319	Open	[VP D4 Board V6 + D401 GMSL]: Tare calibration convergence impact on Auto Exposure

Table 2-2. Specification Changes

Number	Specification Changes
	Firmware v5.10.13 adds support for RealSense™ D435i camera.
	Firmware v5.9.2+ adds USB 2.0 support for RealSense™ D410, D415 and D435 cameras. The USB2.0 is supported for OS Linux and Windows*10 with RealSense™ SDK 2.10.4+ To ensure the best of quality of service, connection to a dedicated USB2 root port is desired.

Summary Table of Changes

Number	Specification Changes
RSDSO-9015	Firmware 5.11.15 allows a D4xx camera to run as an R200 via a JSON file. LibRealSense 2.20.0 or higher is required for this feature.
RSDSO-12258	Firmware 5.11.11.100 and LibrealSense 2.23.0 will bring improved depth linearity and absolute accuracy. Please see Subpixel Linearity Improvement for Intel® RealSense™ Depth Camera D400 Series
RSDSO-12399	Firmware 5.11.11.100 provides USB 2.0 support for D430 module. LibRealSense 2.21.0 or higher is required for this feature.
RSDSO-12266 RSDSO-12267	Firmware 5.11.15.0 provides the ability for Self-Calibration (RealSense™ Self-Calibration) to improve depth noise/precision. Also provides the ability for Tare calibration (RealSense™ Self-Calibration) to improve absolute accuracy. LibRealSense 2.29.0 or higher is required for this feature on D400 series devices.
RSDSO-14282	Firmware 5.12.3.0 provides additional USB2.0 formats for depth and IR streams. The available format for depth and IR 848x480 @ 6FPS, 10FPS.
RSDSO-12490 RSDSO-12281	Firmware 5.12.5.0 provides single frame triggering feature called External Synchronization (GenLock) for D43x global shutter only cameras. The release also includes support for 256x144 @ 300FPS and 848x100 @ 300FPS formats for D43x cameras.
RSDSO-15547 RSDSO-15001 RSDSO-15606	Firmware 05.12.08.200 provides support for depth HDR. For the D455 camera, enable synthetic RGB from left imager and half disparity. Note: Firmware 05.12.08.200 includes a change in API, when using the emitter on/off feature use the latest LibRealSense 2.39.0 or newer with Firmware 05.12.08.200 or newer. Depth HDR requires the latest LibRealSense 2.39.0 or newer with Firmware 05.12.08.200 or newer.
RSDSO-17182	Firmware 5.12.15.50 adds 8FPS for depth+IR1+IR2@848x480 on D435/D430 over USB2
RSDSO-17315	Firmware 5.12.15.50 enables D400 Color AE Priority by default
RSDSO-17425	Firmware 5.12.15.50 adds support for unrectified calibration + RGB YUY2 for D405 required for OEM calibration
RSDSO-18691	D435f and D435if are detected as separate products and can be identified by the SDK and the RealSense viewer starting Firmware 5.15.0.2 and SDK version 2.54.1, from production line.
RSDSO-18278	Firmware 5.15.0.2 adds APIs to enable enhanced Auto Exposure algorithm for D455
RSDSO-19116	Firmware 05.15.1.0 adds support for Manual Depth HDR for D457
RSDSO-19302	Firmware 05.16.0.1 adds support for variable IMU gyro sensitivity on D455 and D435i. check further info here
FW version 05.15.1.55	Firmware version 05.15.1.55 is at production line for selected cameras and is equal to 05.15.1.0 functionality. Refer to PCN #830705-00
FW version 5.16.3.0	Support for a new camera Module D421 with FW ver 5.16.3.0. This FW was only validated on D421, is not available to download and comes pre-loaded on the D421 module.
FW version 5.17.0.9	Firmware 5.17.0.9 supports all the D400 products as part of SDK ver 2.56.4
FW version 5.17.0.10	Firmware 5.17.0.10 supports all the D400 products as part of SDK ver 2.56.5
FW version 5.17.0.213	Firmware 5.17.0.213 supports D436 camera
RSDEV-3823	D436 doesn't support USB2 mode
FW version 5.17.2.201	Firmware 5.17.2.201 supports VP D4 Board V6 + D401 GMSL

3 Errata

3.1 Open

RSDSO-6804 RSDSO-8681	D400 Series camera intermittently enumerated as USB2 device on unplug/plug
Problem:	D400 Series camera intermittently enumerates as a USB 2.0 high speed device when the camera is plugged to a USB 3.1 Gen1 port.
Implication:	The issue is seen on Windows* and Linux*. The issue is not applicable when Host to Camera connection is Type-C (Host) to Type-C (Camera)
Workaround:	Plug in to Host (USB-Type A) after camera connection (Type-C) or alternately physical unplug-plug a camera with different insertion speeds. Issue more likely to occur on slow plug insertion into USB 3.1 Gen1 port.
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10503	Low fill rate in outdoor environment using Auto Exposure (AE)
Problem:	In outdoor light condition (at shaded area), the fill rate might be lower compared to older firmware versions.
Implication:	The failure is observed with Depth Camera D435
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14504	D435i – Frame drops in gyro stream on Windows
Problem:	Start multiple streams (IR, Color, Gyro and Accelerometer), receive frame drops on gyro stream.
Implication:	The failure is observed with Depth Camera D435i
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14455	Frames stop arriving after few minutes on Windows 10 RS4
Problem:	Start streams (Depth, IR, Color, Gyro and Accelerometer), frames stop arriving after a few minutes of streaming.
Implication:	The failure is observed with Depth Camera D400 series
Workaround:	This issue is fixed in Windows 10 RS5
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10591	In Multi Camera mode, sporadic and inconsistent frame drops and streaming halt
Problem:	When Multi-camera configured:

Errata

	- Depth+Color VGAX30FPS - Start-stop streaming, 30 sec streaming duration There is sporadic: - High rate of frame drops received - Exception 'profile not found' received
Implication:	The failure is observed with Depth Camera D400 series
Workaround:	Application should be restarted in order to continue to work
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14308	D415 – Self-Calibration/Tare fails to converge 20% of the time
Problem:	During Self-Calibration and Tare process, the calibration fails to converge 20% of the time.
Implication:	The failure is observed with Depth Camera D415
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-17832	D455 & D415: Focal Length on chip calibration does not work with USB2
Problem:	When selecting Focal Length Calibration - there is no IR ROI preview which is required for adjusting the camera position so the target is inside yellow rectangle of both left & right images
Implication:	On Chip calibration cannot work in USB2 and therefore is blocked for D415 & D455
Workaround:	Switch to USB3
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-19055	D450 + D4 ASIC board with FW version 5.12.7.100 cannot be updated
Problem:	When connecting a D4 ASIC board V3 with FW version <=5.12.13.50 with a new D450 optical module it is not recognized and cannot be updated See PCN 119628-01
Implication:	Cannot build a functional D455 like camera
Workaround:	Connect a different optical module: D410, D415 or D430 and update the FW to a version >= 5.13.0.50.

Status:	Refer the Summary Tables of Changes
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RSDSO-19117	D457 and D456 Aux cap pop up after humidity and temperature stress test
Problem:	After 700 hours of humidity (90%) and temperature (60°) stress test, a few units had the aux cap pop up. There are no reports of this issue occurring with the products under normal operating conditions
Implication:	The unit is not closed and won't meet the IP65 rating
Workaround:	Close the cap
Status:	Refer the Summary Tables of Changes

RSDSO-19280	Self-Calibration: 'NEW' button doesn't show the updated calibration on RealSense Viewer
Problem:	When running Self calibration: On Chip , Tare or Focal calibration types and then toggling between the NEW / ORIGIN buttons (Before initiating 'APPLY NEW' or 'DISMISS'), the updated calibration isn't shown
Implication:	No impact on the self-calibration quality
Workaround:	None
Status:	Refer the Summary Tables of Changes

RSDSO-19114	D45x Z-accuracy out of Spec at high temp
Problem:	D45x cameras don't meet the Z-accuracy spec at 50°C case temperature as found in a 3-temperature qualification test (0°C, 25°C, 50°C)
Implication:	The Z-accuracy and hence depth quality, doesn't meet the KPI
Workaround:	Run self-calibration when the temperature is high or wait for the camera to cool down.
Status:	Refer the Summary Tables of Changes

RSDSO-19017	Up to 80% dropped frames in several builds of Windows 11
Problem:	When streaming at any frame rate and resolution with Windows 11 (KB5035853 OS Builds 22621.3296 and 22631.3296), high frame drop is experienced
Implication:	There's 50% probability to get up to 80% dropped frames
Workaround:	Switch to Windows 10 RS5 or Win 11 KB5030219 (OS Build 22621.2283) where there are no frame drops
Status:	Refer the Summary Tables of Changes

RSDSO-20056	D421: Laser is flickering at low exposure values (<1ms)
Problem:	Laser flickering with AE (due to exposure values < 900us) with D421 module
Implication:	These exposure values are easily reached at minZ range

Errata

Workaround:	Work in manual fixed exposure >1msec
Status:	Refer the Summary Tables of Changes

RSDSO-19836	D421: high-pitched sound heard when streaming depth with laser
Problem:	High-pitched sound heard when streaming depth with laser using D421 module
Implication:	No impact on performance or reliability
Workaround:	Turn laser off, or just ignore the sound
Status:	Refer the Summary Tables of Changes
RSDSO-20061	D455/D457: AE issue when streaming at 90fps with laser off and AE enabled
Problem:	In dark environment with static warm LED light, AE might lock on a certain value when streaming at 90fps with laser off and AE enabled
Implication:	The AE mechanism might lock on a certain value and not change
Workaround:	Avoid this setting of 90FPS and laser off, reset the camera if the scenario occurred
Status:	Refer the Summary Tables of Changes

RSDSO-19865	Laser power change during streaming doesn't work
Problem:	When streaming Depth + IR and changing the laser power while streaming from default 150 to 360, the laser power isn't changed (remains default at 150)
Implication:	The laser power real value is wrong, not as the user set it and there is no increase in the temperature or power consumption
Workaround:	Get the control value after setting the laser power returns the laser power to the real (default) power
Status:	Refer the Summary Tables of Changes

RSDSO-20018	[D45x] Frames gets corrupted while streaming without laser and the lighting changes
Problem:	When streaming with laser turned off and lighting conditions change from light to dark, depth frames might get corrupted
Implication:	Corrupted depth in this condition
Workaround:	Reset the camera (HW rest)
Status:	Refer the Summary Tables of Changes

Status:	Refer the Summary Tables of Changes
RSDEV-3787	[D436] Potential RGB image quality issues
Problem:	Some D436 SKUs show degraded image quality at low light conditions
Implication:	Some D436 SKUs show non uniform dark image and noisy at low lighting
Workaround:	Switch to Manual exposure and shorten the exposure time and gain

Status:	Refer the Summary Tables of Changes
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RSDSO-20529 / RSDSO-16147	[D45x/D436] No sync between color and depth at 5FPS / 6FPS
Problem:	No depth-RGB sync at 5 FPS/6 FPS with D45x/D436 cameras
Implication:	This impacts applications that require the depth and color streams to be synchronized
Workaround:	Set higher FPS such as 15 or 30
Status:	Refer the Summary Tables of Changes

RSDSO-20465 / RSDSO-20572	[D436] Sequential frame drops when streaming color and depth at high resolution @30 FPS
Problem:	High sequential frame drops when streaming color and depth at 1280X720 resolution @30 FPS
Implication:	The issue depends on the platform, seen on Windows NUC and Jetson Xavier
Workaround:	Reduce the resolution/FPS or number of streams
Status:	Refer the Summary Tables of Changes

RSDEV-4648	[D436] Over high temperature there's degradation in the RGB MTF mainly at the corners
Problem:	Over high temperature there's degradation in the RGB MTF mainly at the corners
Implication:	The RGB focus is degraded mainly around the corners at high temperature
Workaround:	Check the datasheet for the camera's assembling guidelines to ensure performance over temperature
Status:	Refer the Summary Tables of Changes

RSDSO-20978	[VP D4 Board V6 + D401 GMSL]: OEM calibration is supported via USB only
Problem:	OEM calibration works only via the USB interface, not the GMSL/FAKRA
Implication:	OEM calibration requires connecting to the USB interface
Workaround:	Use USB interface for OEM calibration
Status:	Refer the Summary Tables of Changes

RSDSO-21319	[VP D4 Board V6 + D401 GMSL]: Tare calibration convergence impact on Auto Exposure
Problem:	Occasionally Tare calibration fails to converge in Auto Exposure mode when performed with VP D4 Board V6 (GMSL/FAKRA) & D401 module. This was observed on NVIDIA Orin & Xavier platforms.
Implication:	When tare fails to converge, the exposure value is stuck at a wrong value
Workaround:	Perform HW reset
Status:	Refer the Summary Tables of Changes

Errata

3.2 Fixed

RSDSO-7194	Windows driver [mksrv.sys] may crash with D400 Series camera in stress testing
Problem:	Windows driver crashes in start- stop streaming iterations. It may take hundreds of start –stop streaming iterations for failure to occur.
Implication:	D400 Series camera fails to be recognized in Windows Device Manager
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-7755	Temporary stream hang observed on disabling Auto Exposure (AE) at 1280X720 resolution after any previous 90 FPS stream.
Problem:	Play any 90 FPS depth or left or right imager stream -> stop -> play 1280x720 resolution -> disable Auto Exposure (AE) -> the stream gets stuck for few seconds.
Implication:	Depth or left and right Imager streams are stuck for a few seconds
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-7798	RGB at 60 FPS may not have the right exposure set when exposure is equal/less than -2
Problem:	Manual exposure with value equal or less than -2 may not result in right exposure.
Implication:	Depth module D415 and Depth cameras D415 and D435 with RGB sensor are affected by this issue
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-7849	ROI based depth streaming immediately after change of IR projector power may result in a stream hang
Problem:	Frames do not arrive after ROI (Region of Interest) is selected to start streaming immediately after a change is made to the IR projector power.
Implication:	No depth streaming
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-7854	Depth Stream hang when system resumes from Sleep (S3)
Problem:	System resume from S3 does not resume depth streaming and requires application restart.
Implication:	Currently seen on Windows* only.
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-7976	D400 series camera is not recognized after reboot on Linux
Problem:	Device does not appear in the device manager (lsusb)

Implication:	The frequency of the problem occurrence depends on specific Kernel version. It occurs more frequently on 4.4.0.x kernel versions and less frequently with 4.10.x kernel versions. Not seen on Windows*
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-8007	Firmware updates via DFU Service fails when firmware update limit is reached
Problem:	D400 series firmware update engine will allow a return to a previous version or baseline version of firmware up to 20 times unless a higher version of firmware. DFU service as part of Windows Driver package updates camera firmware when camera connected has a firmware version different than expected. The DFU service fails to function when firmware update limit of 20 is reached.
Implication:	When the firmware update limit is reached, firmware update fails even if higher firmware version. DFU service is in Windows driver package only.
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-8328	Metadata attribute “Trigger” indicating Depth to Color synchronization may not have correct value
Problem:	Trigger is a metadata field and its value indicates whether the depth and color streams are synced (1) or not (0). The value in this metadata field indicating synchronization may have the wrong value.
Implication:	D400 series cameras, D415 and D435 with color sensor
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-8461	D400 Series Windows driver does not work with Chrome browser
Problem:	When Windows driver is installed on a Windows*10 system, chrome browser does not recognize D400 Series camera in chrome://settings/content/camera
Implication:	D400 series camera is recognized without Windows driver installed.
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-8467	Left Imager UYVY format displays green image
Problem:	Streaming color out of left imager in UYVY format displays a green image
Implication:	RealSense Viewer displays a green image when UYVY format is selected for left imager stream
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-8538	Color correction parameters are not updated correctly
Problem:	Color correction parameters update to default values
Implication:	This issue affects color from left imager in cameras D400, D410 & D415
Status:	Refer the <i>Summary Tables of Changes</i>

Errata

RSDSO-8565	Infrared speckles on color image from D415 and D435 cameras
Problem:	Infrared speckles are seen on color image from D415 and D435 cameras when laser power is at maximum or closer to maximum value
Implication:	Infrared speckles reduces with distance and ambient lighting
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9006	Frame rate does not change when manual exposure value is changed
Problem:	Frame rate (FPS) may need to change based on the exposure value and in some cases the FPS may not change as expected.
Implication:	Issue observed with camera D430 and D435
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9074	Simultaneous streaming Depth, Imager and Color may result in data stream hang when camera is connected through USB2
Problem:	One or two streams hangs may hang when simultaneously streaming Depth, Imager and Color data when camera is connected through USB2
Implication:	The issue is not observed when 1 or 2 data streams are simultaneously streaming. The issue is observed on Windows* and Linux*
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9094	Specific controls values missing in frames metadata
Problem:	Frames arrive without controls values in Metadata
Implication:	Missing metadata for valid frames. Issue observed in Linux*
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9153	D400 series camera fails to be recognized on system reboot when connected through USB3
Problem:	D400 Series camera may fail to be recognized on system reboot when connected through USB3
Implication:	The issue is observed on Windows*. Camera not recognized in Windows Device Manager
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9224	IR Projector pattern flicker when streaming through USB2 connection
Problem:	IR Projector pattern flicker maybe observed when camera is streaming through a USB2 connection
Implication:	The flicker may be observed after streaming for some time (~3 minutes) independent of resolution and frame rate. It is observed in D400 series cameras with IR projectors and on Windows* and Linux*
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9228	D400 series camera disconnects on resume from system sleep when connected through USB2
Problem:	D400 series camera may disconnect on resume from system sleep when connected through USB2.
Implication:	Application such as RealSense Viewer streaming before entering system sleep fail to function on resume from sleep as the camera may fail to be recognized. The issue is only observed on Linux* OS
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9240	D400 Series camera fails to be recognized on system reboot when connected through USB2
Problem:	D400 Series camera may fail to be recognized on system reboot when connected through USB2
Implication:	The issue is observed on Windows* and Linux*
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9478	Image Flicker when Auto Exposure (AE) is enabled
Problem:	Image flicker may be observed under certain light conditions when Auto Exposure (AE) is enabled
Implication:	Image flicker seen on imager output streams may impact the depth stream
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9546	IR projector pattern flicker when streaming at 1280X720, 4 FPS and connected through USB2
Problem:	IR projector pattern flicker may be observed when streaming at resolution 1280X720, 4 FPS and camera connected to a USB2 connection
Implication:	The issue is observed on Windows* and Linux*
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9556	Camera stuck after streaming start-stop at Low FPS for few times
Problem:	Camera depth streams at low frame rates may be stuck after start-stop streaming a few times
Implication:	The issue is observed for both, USB3 and USB2 camera connection and at 6FPS. Re-plugging the camera is required to be able to communicate with the camera again.
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9645	Darker depth frame when changing depth exposure from [165760 - 165780] and connected through USB2
Problem:	Darker (holes) depth frames are observed when depth exposure is changed between 165760 and 165780 range of values
Implication:	When streaming depth / IR configuration with resolutions [480x270/640x480] and [6/15/30/ 60] fps and camera connected through USB2
Status:	Refer the <i>Summary Tables of Changes</i>

Errata

RSDSO-10002	Calibration tables may get corrupted during power on and off cycles
Problem:	Power on/off cycles may cause calibration table to get corrupted
Implication:	Invalid depth stream
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10011	Auto Exposure (AE) for Color is not optimized for bright sunlight
Problem:	Auto Exposure for Color is not optimized for bright sunlight.
Implication:	Depth Cameras D415 and D435 support color through dedicated RGB sensor and are impacted by this issue. The cameras cannot be used for color in bright sunlight for use cases that require good AE based region of interest (ROI)
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10428	IR Image is black in Auto Exposure (AE) mode with sudden exposure to light
Problem:	When switching from no light to full outdoor sunlight in AE mode, IR image turns black
Implication:	The failure is observed with Depth Camera D415
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10431	IR image may flicker in outdoor sunlight when using Auto Exposure (AE) with default set point
Problem:	IR image flickers in outdoor sunlight when using Auto Exposure with default AE set-point (1536) and higher
Implication:	The failure is observed with Depth Camera D435
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10674	D435i – The first (cold) start of IMU sensors in LibRealSense on Linux takes ~4 sec
Problem:	Sending request to stream Accel/Gyro data results in 4 sec wait till the data starts to arrive.
Implication:	The failure is observed with Depth Camera D435i
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10777	D435i - Buffer overflow on repeated start/stop
Problem:	Cycling through start/stop with Depth+Gyro+Accelerator streams abruptly terminates with: Process finished with exit code 134 (interrupted by signal 6: SIGABRT)
Implication:	The failure is observed with Depth Camera D435i
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-11040	D435 – Depth/IR corrupted image when streaming multi stream and RGB exposure is < -6
Problem:	Depth/IR corrupted image when streaming multi stream and RGB exposure is less than -6
Implication:	The failure is observed with Depth Camera D435
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-11041	D435i – Unreasonably large accelerometer reading in Windows 10
Problem:	Unreasonably large accelerometer reading in Windows 10
Implication:	The failure is observed with Depth Camera D435i
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-11042	D430 - RealSenseViewer crash after win10 PC reboot
Problem:	RealSense Viewer errors out post reboot after using the “Hand” preset
Implication:	The failure is observed with D415, -435, -430
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-12586	D415, D435/D435i – RGB camera not available after FW Update process using DFU
Problem:	Upgrade device using DFU after upgrade process is complete, RGB camera not available
Implication:	The failure is observed with D435/D435i and D415
Workaround:	Do not disconnect USB cable for 20+ secs after FW update process is complete. This allows the host system to enumerate the device properly. If device is experiencing this issue, please use the following steps to downgrade FW to recover Downgrade device FW to FW 5.11.4 using DFU tool (making sure to do not disconnect USB cable for 20+ secs after FW update is complete and device has been enumerated and detected by host platform)
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-12587 RSDSO-12814	D435i – Upgrade to latest Major FW causes corrupted calibration table
Problem:	Upgrade device to latest Major FW causes corrupted calibration table
Implication:	The failure is observed with D435i
Workaround:	If the device has been upgraded to FW 5.11.6.200, run LibRealSense ver 2.23 (or higher), and it will correct the calibration table. Alternatively, downgrade FW to FW 5.10.13 and verify that the calibration table is OK.
Status:	Refer the <i>Summary Tables of Changes</i>

Errata

RSDSO-10229 RSDSO-13386	Camera fail after start, stop of random profiles
Problem:	When trying to stream more than one camera stream (depth, infrared and color) in a random profile, the camera streams aborts.
Implication:	None
Workaround:	Disconnect and reconnect camera
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-13540	D415 –RealSense™ Self-Calibration doesn't converge using D415 camera
Problem:	When running RealSense™ Self-Calibration process via LibRealSense, the error "Calibration didn't converge! (EDGE_TO_CLOSE) please retry in different lighting conditions" is seen.
Implication:	The failure is observed with Depth Camera D415
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-13554	D420 – Camera will stop working after FW update
Problem:	When upgrading FW, the D420 stops working.
Implication:	The failure is observed with Depth Module D420
Workaround:	If D420 is not experiencing issue, do not upgrade FW
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-13546	D435i – IMU frame drops
Problem:	When depth, infrared, color, gyro and accel are streaming for a long period of time, frame drops (gyro and accel) are observed.
Implication:	The failure is observed with Depth Camera D435i
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10920	First frames not received on metadata test
Problem:	Metadata start frame on depth/IR/color stream began from 2 or 3 value, the expected value is 1.
Implication:	None
Workaround:	Ignore first frame of metadata
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-12578	Device not recognized in device manager after machine reboot
Problem:	When rebooting system under Windows operating system, camera is not recognized by device manager.
Implication:	None
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14309	Scale error after Tare calibration (RealSense™ Self-Calibration) was significantly degraded
Problem:	When using Tare calibration functionality, the scale error can be significantly degraded.
Implication:	None
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-9501	Camera is not functional after HLK Sensor test when connected through USB2
Problem:	Camera is not functional after HLK Sensor test when connected through USB2 (Windows HLK)
Implication:	The issue is observed on production units and not seen on pre-production samples.
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-10603	Unable to set the Depth Exposure Time < 70 usec through USB2
Problem:	When connected through USB2, setting Depth Exposure Time less than 70 usec will result in artifacts and over-exposed frames.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14526	D435i – Sending HW_reset immediately after stop streaming causes the camera to boot into invalid state
Problem:	Sending HW_reset immediately after stop streaming causes the camera to boot into invalid state.
Implication:	The failure is observed with Depth Camera D435i
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14517	10 FPS not supported on Depth/IR 848x480 when connected through USB2
Problem:	When connected through USB2, 10 FPS is not available on depth or IR streams under 848x480 resolution.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

Errata

RSDSO-14337	IR Frames stop arriving after several camera start/stop
Problem:	During several iterations of start of frame stream, collect frames and stop streaming, the IR frames stop arriving.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-13306	Frames stop arriving after few frames in Color/Depth stream due to unexpected halt on Windows 10 RS4
Problem:	Frames stop arriving after few frames in Color/Depth stream due to unexpected halt on Windows 10 RS4.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14558	Fails USB enumeration during PC restart
Problem:	Camera sometimes fails USB3.1 enumeration during PC restart.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14512	D435i – Failed to start IMU stream after resetting device
Problem:	IMU stream fails to start after camera reset command is sent to the device.
Implication:	The failure is observed with Depth Camera D435i
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14349	FW Update ID mismatch before and after entering DFU
Problem:	Before and after entering DFU for FW Update, the FW Update ID shows mismatch.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14525	D435i – Camera not released to idle state on Ubuntu 18.04 with Kernel 5
Problem:	D435i camera not going into idle state on Ubuntu 18.04 with Kernel 5.
Implication:	The failure is observed with Depth Camera D435i
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14499	Config error when stopping Color stream on Windows 10 RS4
Problem:	With camera connected and RealSense Viewer open, starting Color stream (exposure min value), then changing exposure to max value and stopping Color stream causes error.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-16289	When laser emitter is enabled via Python, laser turns on and off continuously
Problem:	When setting laser emitter to enabled in Python, laser turns on and off continuously.
Implication:	The failure is observed with Depth Camera D400 series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-16765	Frames not arrived with 300fps
Problem:	When setting depth 256x144 @ 300FPS, frames not arrived.
Implication:	The failure is observed with Depth Camera D400 Series
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14530	Frame drops when running on Ubuntu 18.04 with Kernel 5
Problem:	Start multiple streams (IR, Color, Gyro and Accelerometer), receive frame drops on streams.
Implication:	The failure is observed with Depth Camera D400 series
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-14978	D455 – Depth/IR frame drops
Problem:	When setting depth and IR resolution 480x270 @ 5FPS, can take up to 8 sec before first frame arrives.
Implication:	The failure is observed with Depth Camera D455
Workaround:	None
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-18928	New D455 cameras can't downgrade FW to <= 5.12.15.50
Problem:	See PCN 119628-01
Implication:	Any attempt to downgrade the firmware to version 5.12.13.50 and lower will lock the camera. Downgrades to other firmware versions will cause the camera to enter a Device Firmware Update mode, which still allows for the firmware version to be updated.
Workaround:	The firmware version of the cameras and modules with the new sensors can be updated to version number 5.13.0.50/5.13.0.55 or higher.

Errata

Status:	Refer the <i>Summary Tables of Changes</i>
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RSDSO-19297	D457 camera in recovery mode cannot be recognized by the Viewer
Problem:	Start updating the FW in the D457 camera and interrupt the process so the camera enters recovery mode. In this mode the RealSense Viewer cannot detect the camera.
Implication:	The camera's FW cannot be updated and return to active mode
Workaround:	Copy the FW image to the host and copy it to the camera via command line
Status:	Refer the Summary Tables of Changes

RSDSO-17488	D405 High UV error when aspect ratio is different than 16/9 (HD)
Problem:	When streaming high resolution color at aspect ratio different than 16/9 there are high number of UV mapping pixel errors
Implication:	Significant degradation of more than ~15 pixel when using resolution with aspect ratio of 4/3 (VGA) in D405
Workaround:	Use 16/9 aspect ratio
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-19674	Corrupted depth and IR frames in start stop scenario
Problem:	In random streaming scenarios the depth and IR frames are corrupted. This can occur after start/stop, or streaming of Depth, IR and RGB, when the projector can be turned On or Off.
Implication:	When the projector is Off the frames are black. When the projector is On, the upper part of the depth is corrupted, and the bottom part is black.
Workaround:	Stop and restart the stream
Status:	Refer the Summary Tables of Changes

RSDSO-19269	D435 RGB enumeration failure after a few HW reset
Problem:	When sending a few consequent HW reset commands from the Viewer (or even just physically disconnect/connect the device), sometimes the RGB sensor doesn't enumerate properly. This issue is reproduced with D435 only
Implication:	The RGB sensor cannot stream from the viewer, though it exists in the device manager
Workaround:	Close and re-open the Viewer, or wait for 0.5 sec and re-enumerate the RGB sensor
Status:	Refer the Summary Tables of Changes

RSDSO-14986	D455/D457 HW timestamp delta is 4.8FPS and not 5FP
Problem:	D455/D457 HW timestamp delta is 4.8FPS and not 5FPS
Implication:	The HW TS delta is often higher than expected (208ms - 218ms) while the expected HW TS delta between two sequential frames with 5 fps is 200ms

Workaround:	Change the FPS
Status:	Refer the <i>Summary Tables of Changes</i>

RSDSO-20426	Corrupted frames when streaming @ 60 FPS in AE - D455x and D456
Problem:	When streaming Depth+RGB at 60FPS in auto exposure mode the Depth frames are corrupted
Implication:	Depth frames are corrupt
Workaround:	Set Depth or RGB FPS to 5, 15, 30
Status:	Refer the Summary Tables of Changes

RSDSO-20384	[D45x] RGB Manual exposure at 60FPS doesn't work at certain range
Problem:	When setting RGB to 60FPS and manual exposure <=156 (unit of 100 nanoseconds) the frames are black
Implication:	Streaming RGB at 60FPS and manual exposure <=156 doesn't work
Workaround:	Change RGB setting to Auto Exposure, or set FPS to 5, 15, 30 or to Manual exposure > 156